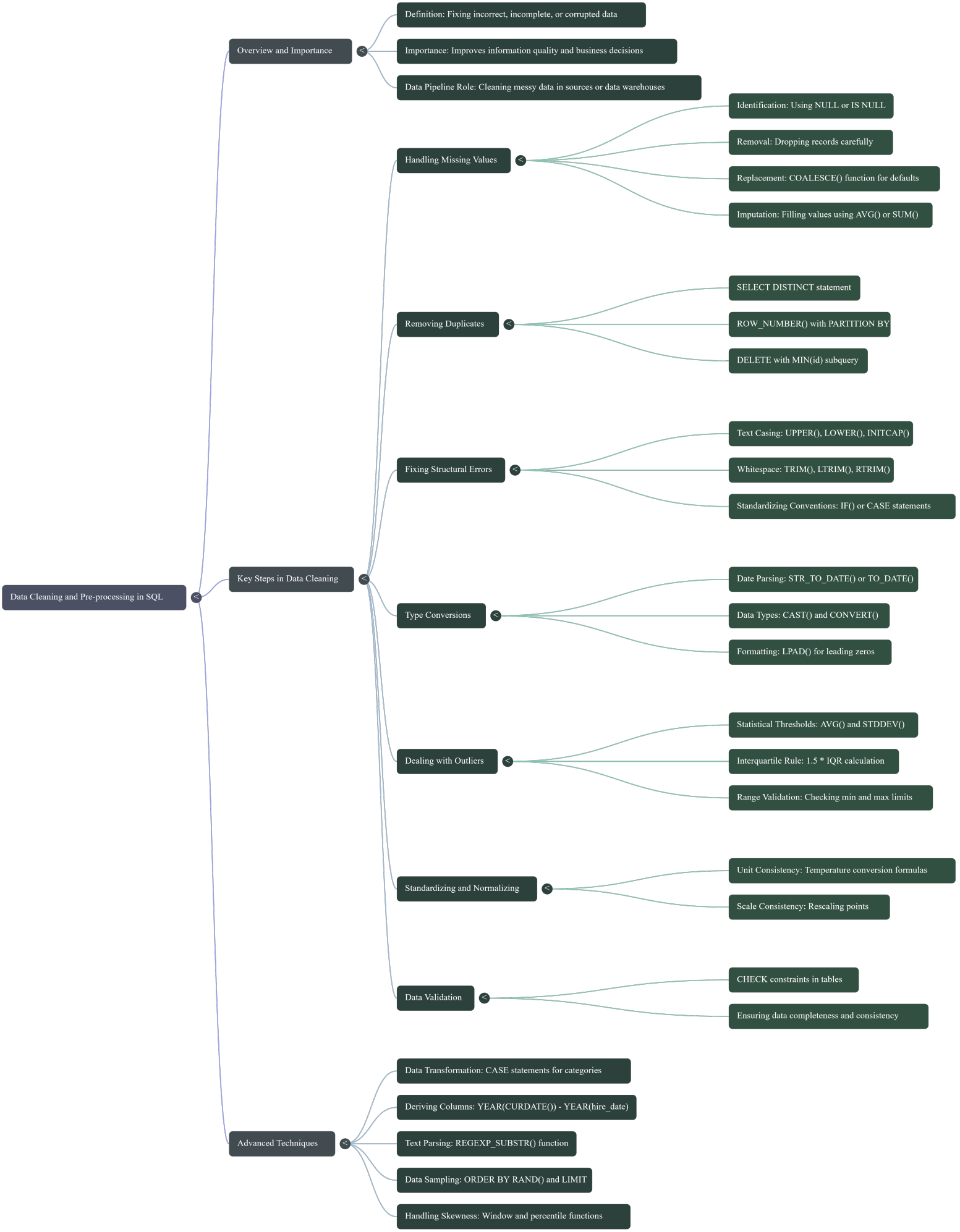

Data Cleaning and Pre-processing in SQL
Overview and Importance
Definition: Fixing incorrect, incomplete, or corrupted data
Importance: Improves information quality and business decisions
Data Pipeline Role: Cleaning messy data in sources or data warehouses
Key Steps in Data Cleaning
Handling Missing Values
Identification: Using NULL or IS NULL
Removal: Dropping records carefully
Replacement: COALESCE() function for defaults
Imputation: Filling values using AVG() or SUM()
Removing Duplicates
SELECT DISTINCT statement
ROW_NUMBER() with PARTITION BY
DELETE with MIN(id) subquery
Fixing Structural Errors
Text Casing: UPPER(), LOWER(), INITCAP()
Whitespace: TRIM(), LTRIM(), RTRIM()
Standardizing Conventions: IF() or CASE statements
Type Conversions
Date Parsing: STR_TO_DATE() or TO_DATE()
Data Types: CAST() and CONVERT()
Formatting: LPAD() for leading zeros
Dealing with Outliers
Statistical Thresholds: AVG() and STDDEV()
Interquartile Rule: 1.5 * IQR calculation
Range Validation: Checking min and max limits
Standardizing and Normalizing
Unit Consistency: Temperature conversion formulas
Scale Consistency: Rescaling points
Data Validation
CHECK constraints in tables
Ensuring data completeness and consistency
Advanced Techniques
Data Transformation: CASE statements for categories
Deriving Columns: YEAR(CURDATE()) - YEAR(hire_date)
Text Parsing: REGEXP_SUBSTR() function
Data Sampling: ORDER BY RAND() and LIMIT
Handling Skewness: Window and percentile functions